

AIDataCenter

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Contents

1 Contents	3
2 Highlights	4
3 Quick start	5
4 Documentation (PDF)	6
5 License	7

Specification and bootstrap stack for a **64× NVIDIA B200** research AI inference cluster.

1 Contents

Path	Description
SPEC.md	Full facility / fabric / storage / K8s design
docs/	Build guide, network (ZTP/RoCE), overlay (EVPN/VXLAN), serving (Kimi K2 + API), bootstrap, NetBox SoT, cabling guide
bootstrap/	Seed host, Metal3/CAPI, Flux platform, NetBox
demo/	Interactive live bring-up simulator (browser)

2 Highlights

- **4 racks:** GPU-1, GPU-2, BOOT, STOR
- **Fabric:** Arista 7060DX5 rail-optimized 8×400G/node (RoCEv2)
- **OOB:** Arista 7010TX-48
- **Platform:** Kubernetes + NVIDIA GPU Operator + KServe + Ray
- **Serving:** Kimi K2 Thinking (TP=8 × replicas) behind Envoy gateway + inference scheduler ([docs/serving.md](#))
- **Overlay:** EVPN/VXLAN — vrf storage + vrf edge for API ingress ([docs/overlay.md](#))
- **Source of truth:** [NetBox](#) for inventory, IPAM, and cabling
- **GitOps:** Flux-managed platform apps (Harbor, Vault, observability, Redpanda, ...)

3 Quick start

```
# Interactive demo (simulated bootstrap → switches → GPU cluster)
python3 -m http.server 8765 --directory demo
# open http://127.0.0.1:8765

# Design-time inventory + cabling (no hardware required)
bash bootstrap/scripts/netbox-sync.sh --offline

# Physical build: docs/build-guide.md → then docs/bootstrap.md + bootstrap/README.md
```

4 Documentation (PDF)

Every Markdown doc has a Typst twin + compiled PDF next to it (e.g. `docs/build-guide.typ` → `docs/build-guide.pdf`, `README.pdf`). Regenerate all after editing any `.md` source:

```
pip install pypandoc-binary # plus typst CLI on PATH
python3 docs/md2pdf.py      # or: python3 docs/md2pdf.py docs/build-guide.md
```

Styling is shared via `docs/template.typ`; generated `.typ` files are overwritten on each run — edit the `.md` or the template, not the generated output.

5 License

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